Laboratory Report # 2

Name: *Cyril Andre B. Durango* **Date Completed:** *September 6, 2024*

Laboratory Exercise Title: *Basic Constructs in Verilog HDL*

Target Course Outcomes:

CO1: Create descriptions of digital hardware components such as in combinational and sequential circuits using synthesizable Verilog HDL constructs.

CO2: Verify the functionality of HDL-based components through design verification tools.

Exercise 2C: 4-bit Adder. Using the full adder design in Exercise 2B, create a structural description of a 4-bit adder. Use the entity and structural diagrams below for reference. Synthesize the design**.**

A four-bit adder is created using 4 full-adder circuits connected in series. A full-adder is a circuit that takes in three input signals and adds them, therefore generating two output signals called the “sum” and the “carry”.

INPUTS			OUTPUTS	
x	**y**	**z**	**C**	**S**
0	0	0	**0**	**0**
0	0	1	**0**	**1**
0	1	0	**0**	**1**
0	1	1	**1**	**0**
1	0	0	**0**	**1**
1	0	1	**1**	**0**
1	1	0	**1**	**0**
1	1	1	**1**	**1**

Table 1. Full-Adder Circuit Truth table

x \ yz	00	01	11	10
0	0	0	1	0
1	0	1	1	1

(A)

x \ yz	00	01	11	10
0	0	1	0	1
1	1	0	1	0

(B)

Figure 1. K-Map of the Full Adder Outputs. (A) Carry or C output. (B) Sum or S output

$C = xy + xz + yz$
$S = x \oplus y \oplus z$

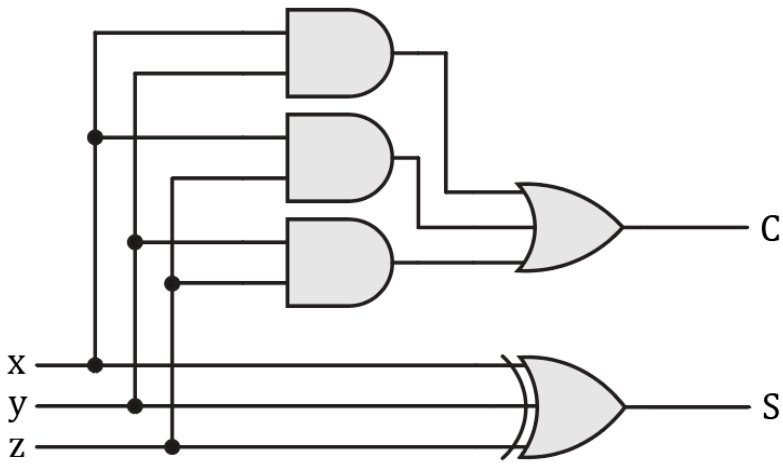
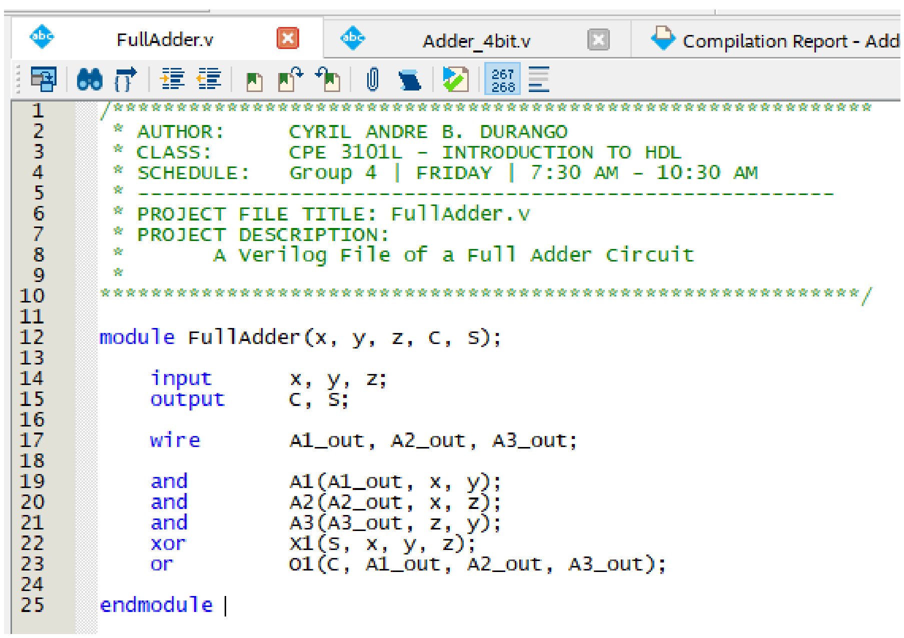


Figure 2. Circuit Diagram of a Full Adder Circuit

Using the equations that are derived from the K-maps, the gate primitives in Verilog can be utilized to simulate the logic behind the outputs.




```
/*****************************************************************
 * AUTHOR:      CYRIL ANDRE B. DURANGO
 * CLASS:       CPE 3101L - INTRODUCTION TO HDL
 * SCHEDULE:    Group 4 | FRIDAY | 7:30 AM - 10:30 AM
 * ---------------------------------------------------------
 * PROJECT FILE TITLE: FullAdder.v
 * PROJECT DESCRIPTION:
 *        A Verilog File of a Full Adder Circuit
 *
*****************************************************************/

module FullAdder(x, y, z, C, S);

    input       x, y, z;
    output      C, S;

    wire        A1_out, A2_out, A3_out;

    and         A1(A1_out, x, y);
    and         A2(A2_out, x, z);
    and         A3(A3_out, z, y);
    xor         X1(S, x, y, z);
    or          O1(C, A1_out, A2_out, A3_out);

endmodule
```

Figure 3. Verilog HDL File simulating a Full-Adder Circuit

Below is the four-bit adder circuit in Verilog, utilizing FullAdder.v in using four full adders.

Adder_4bit.v | tb_Adder_4bit.v

```
/**************************************************************
 * AUTHOR:     CYRIL ANDRE B. DURANGO
 * CLASS:      CPE 3101L - INTRODUCTION TO HDL
 * SCHEDULE:   Group 4 | FRIDAY | 7:30 AM - 10:30 AM
 * -------------------------------------------------------
 * PROJECT FILE TITLE: Adder_4bit.v
 * PROJECT DESCRIPTION:
 *       A Verilog File of a Four-bit Adder Circuit
 *
 *************************************************************/

module Adder_4bit(X,Y,Z,C,S);

   input     [3:0] X, Y;
   input           Z;

   output    [3:0] S;
   output          C;

   wire      [2:0] Carry_out;

   FullAdder bit_1(X[0], Y[0], Z,            Carry_out[0],  S[0]);
   FullAdder bit_2(X[1], Y[1], Carry_out[0], Carry_out[1],  S[1]);
   FullAdder bit_3(X[2], Y[2], Carry_out[1], Carry_out[2],  S[2]);
   FullAdder bit_4(X[3], Y[3], Carry_out[2], C,             S[3]);

endmodule
```

Figure 4. Verilog HDL File simulating a Four-bit Adder Circuit

Adder_4bit.v | Compilation Report - Adder_4bit

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Flow Summary

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Flow Status	Successful - Fri Sep 13 09:53:20 2024
Quartus Prime Version	20.1.1 Build 720 11/11/2020 SJ Lite Edition
Revision Name	Adder_4bit
Top-level Entity Name	Adder_4bit
Family	MAX 10
Device	10M02DCU324A6G
Timing Models	Final
Total logic elements	6
Total registers	0
Total pins	14
Total virtual pins	0
Total memory bits	0
Embedded Multiplier 9-bit elements	0
Total PLLs	0
UFM blocks	0
ADC blocks	0

Figure 5. Adder_4bit. v Flow Summary After Compilation

The RTL Viewer of the 4-Bit Adder Circuit utilizes four full-adder circuits as theorized, and each of the full adder circuits contains the same logic circuit diagram illustrated above.

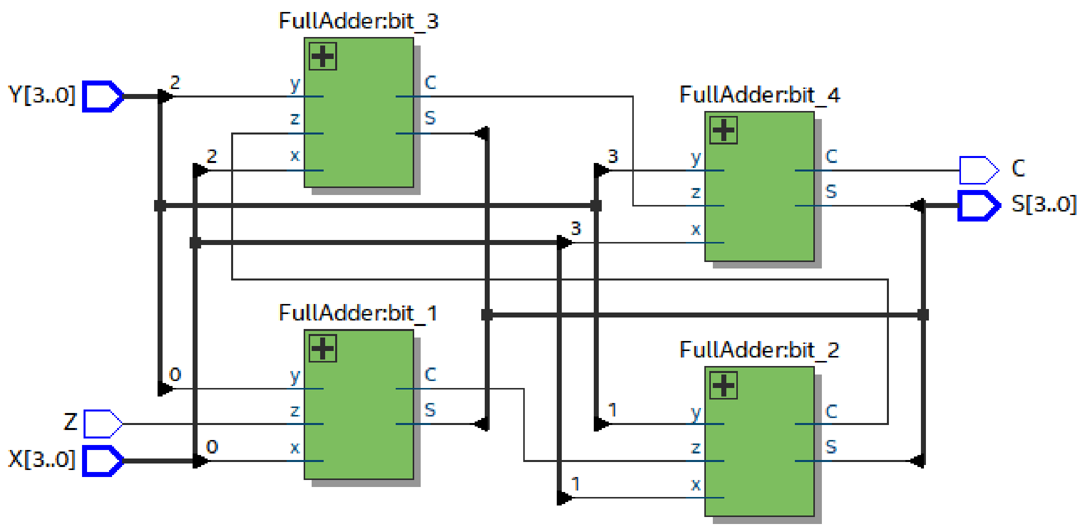

Figure 6. RTL Block Diagram View of the Four-bit Adder Circuit.

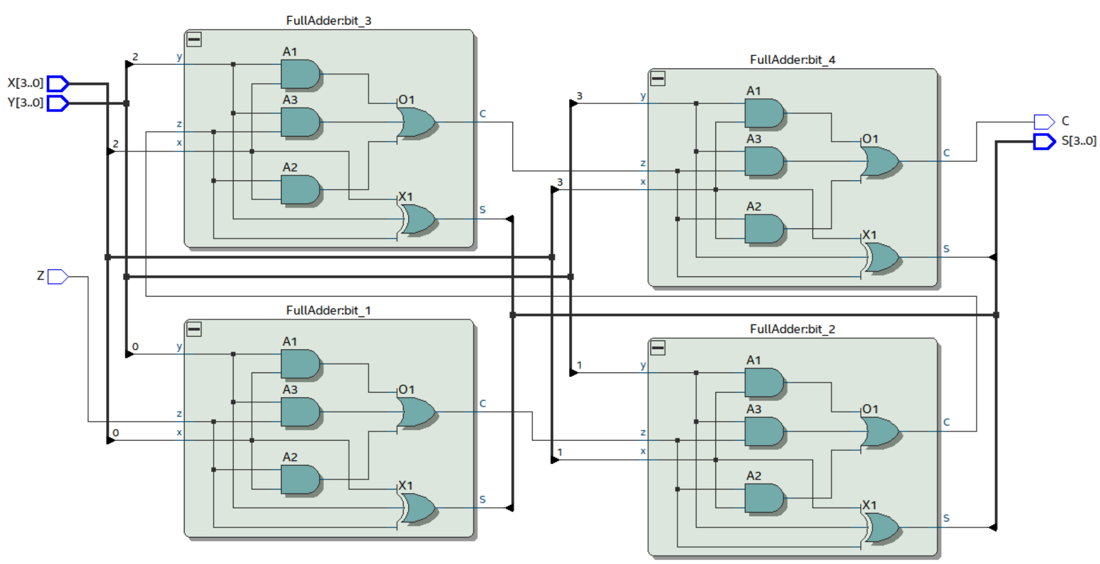

Figure 6. RTL Circuit Diagram of the Four-bit Adder Circuit.

Below is the testbench file for the Four-bit adder circuit with several test values. Following that is the waveform of these values and their expected results

```
/*************************************************************
 * AUTHOR:     CYRIL ANDRE B. DURANGO
 * CLASS:      CPE 3101L - INTRODUCTION TO HDL
 * SCHEDULE:   Group 4 | FRIDAY | 7:30 AM - 10:30 AM
 * -------------------------------------------------------
 * PROJECT FILE TITLE: Adder_4bit.v
 * PROJECT DESCRIPTION:
 *       A Testbench Verilog File of "Adder_4bit" Circuit
 *
*************************************************************/

`timescale 1 ns / 1 ps
module tb_Adder_4bit();

   reg   [3:0] X, Y;
   reg         Z;

   wire  [3:0] S;
   wire        C;

   Adder_4bit UUT(X,Y,Z,C,S);

   initial
   begin
      X = 4'd0;   Y = 4'd0;   Z = 0; #10
      X = 4'd3;   Y = 4'd8;   Z = 1; #10
      X = 4'd11;  Y = 4'd3;   Z = 0; #10
      X = 4'd12;  Y = 4'd6;   Z = 0; #10
      X = 4'd5;   Y = 4'd4;   Z = 1; #10
      X = 4'd1;   Y = 4'd9;   Z = 0; #10
      X = 4'd15;  Y = 4'd15;  Z = 0; #10
      X = 4'd15;  Y = 4'd15;  Z = 1; #10

      $stop;
   end

endmodule
```

Figure 7. Testbench File for the Four-bit Adder

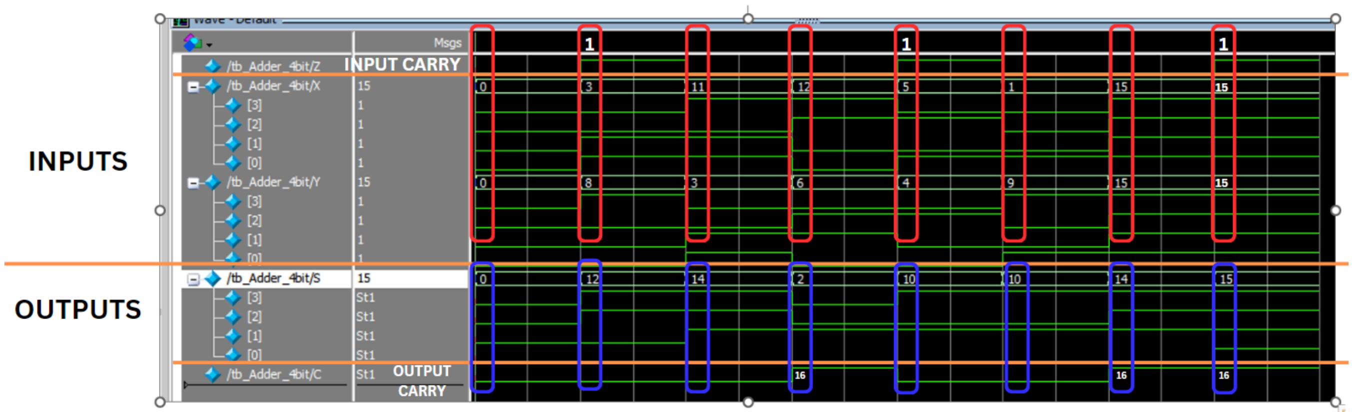

Figure 8. Four-bit Adder Generated Waveform

The main orange line divides the graphed waveforms into two: the inputs and the outputs. Variable Z is the input carry bit, and variable C is the output carry bit. The red and blue rectangles relate to the signals or columns in which they are aligned. The sum of the numerical values found in the red rectangle must equal the sum of the numerical values of the blue rectangle.

Example: in the second column of signals. The red rectangle contains a carry of 1, and the binary representation of 3 and 8. Summing them up together, we get 12. In the blue rectangle directly below the said red rectangle, it displays the binary representation of “12”

Additionally, the output carry bit represents 16 in decimal form, as must at the leftmost part of the summed bits. This is evident in the fourth, seventh, and eighth columns, wherein the sum of the red rectangles (18, 30, and 31 respectively) is more than a typical four-bit output can handle. The output carry bit is the 5th bit of the binary sequence that allows the users to receive a displayed sum of 18 in the fourth column, 30 in the seventh, and 31 in the eighth.